

Dynamic-Equation and Analysis Contract for an IEEE 14-Bus Mixed-Resource System

One EMF6 Synchronous Generator, One 13-State GFM–VSG, and Three 10-State GFL Converters
Power Flow, Equilibrium, Full-State SSSA, Fault-Only and SG-Trip Time-Domain Studies

In-house Power-Flow Project

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Abstract

This report documents the exact positive-sequence RMS equations used by the project for the IEEE 14-bus mixed-resource case. The normal profile contains SG1 at bus 1, IBR2 in 13-state grid-forming (GFM) mode at bus 2, and IBR3, IBR6, and IBR8 in 10-state PLL-resolved grid-following (GFL) mode. The report defines every state, reference frame, sign convention, per-unit conversion, network residual, equilibrium condition, SSSA reduction, and implicit trapezoidal time step. Power flow, SSSA, fault-only TS, and SG trip/return products are executed independently; the launcher “Full Analysis” product is intentionally excluded. Numerical plots are rendered natively by $\LaTeX$ from CSV files exported from the project-owned MATLAB solvers, so the plot font, size, and spacing are identical to the report. Numerical instability or fail-closed outcomes are reported rather than tuned away.

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1 Scope, evidence discipline, and reproducibility

The production calculation uses only project-owned base-MATLAB code. The allowed numerical primitives include `\` and `eig`; no external power-flow, optimization, Simulink, or loaded-solution result feeds the calculation. The evidence files in `\datadir` are generated by `scripts/reporting/generate_ieee14_ibr_pf_sssa_ts_report`. That script calls the same production dispatch used by `solve_case`, writes the full-precision results to CSV and MAT files, and records branch, commit, profile, convergence, and physical classifications in `manifest.txt`.

The products in this report are deliberately separated:

1. normal network power flow (PF), with no dynamic states and no event;
2. mixed-device equilibrium and full-state SSSA at the same fixed mode map;
3. a balanced positive-sequence fault-only TS, with no SG trip;
4. an SG-trip TS with automatic GFM selection, with no network fault;
5. independent pre-trip, SG-tripped, and SG-returned PF/SSSA operating-point comparisons.

The last item is not a time trajectory and its eigenvalue rows are not cross-operating-point mode matches.

2 Case data and index maps

2.1 System and resource map

The network is the 100-MVA, 60-Hz IEEE 14-bus data set. The external bus ID is not assumed equal to an arbitrary internal index, even though this case happens to use consecutive IDs. Algebraic voltage order is

$$\mathbf{y} = [V_{x,1}, V_{y,1}, V_{x,2}, V_{y,2}, \dots, V_{x,14}, V_{y,14}]^T, \quad V_k = V_{x,k} + jV_{y,k}.$$

The fixed normal resource map is

Device index	Bus ID	Device	Normal mode	Physical states	Active in normal SSSA
1	1	SG1	synchronous EMF6	6	5
2	2	IBR2	GFM REGFM_B1	13 of dual 23	13
3	3	IBR3	GFL-RMS10	10 of dual 23	10
4	6	IBR6	GFL-RMS10	10 of dual 23	10
5	8	IBR8	GFL-RMS10	10 of dual 23	10
Active dynamic order				$5 + 13 + 3(10) = 48$	

The total fixed inventory is 98 states: SG1 contributes 6 and each dual IBR contains both the 13-state GFM and 10-state GFL branches. Only the branch selected by the operating mode enters the reduced dynamic matrix.

2.2 Bus and line input data

Table 1: Case bus data. These are inputs, not solved PF results.

Bus	Type	P_L MW	Q_L MVar	V_0 pu	θ_0 deg	Base kV	V_{max}	V_{min}
1	REF	0	0	1.06	0	0	1.06	0.94
2	PV	21.7	12.7	1.045	-4.98	0	1.06	0.94
3	PV	94.2	19	1.01	-12.72	0	1.06	0.94
4	PQ	47.8	-3.9	1.019	-10.33	0	1.06	0.94
5	PQ	7.6	1.6	1.02	-8.78	0	1.06	0.94
6	PV	11.2	7.5	1.07	-14.22	0	1.06	0.94
7	PQ	0	0	1.062	-13.37	0	1.06	0.94
8	PV	0	0	1.09	-13.36	0	1.06	0.94
9	PQ	29.5	16.6	1.056	-14.94	0	1.06	0.94
10	PQ	9	5.8	1.051	-15.1	0	1.06	0.94
11	PQ	3.5	1.8	1.057	-14.79	0	1.06	0.94
12	PQ	6.1	1.6	1.055	-15.07	0	1.06	0.94
13	PQ	13.5	5.8	1.05	-15.16	0	1.06	0.94
14	PQ	14.9	5	1.036	-16.04	0	1.06	0.94

Table 2: Case branch data in project index order.

Index	From	To	r pu	x pu	b_c pu	Rate A	Tap	Shift deg	Status
1	1	2	0.01938	0.05917	0.0528	0	0	0	1
2	1	5	0.05403	0.22304	0.0492	0	0	0	1
3	2	3	0.04699	0.19797	0.0438	0	0	0	1
4	2	4	0.05811	0.17632	0.034	0	0	0	1
5	2	5	0.05695	0.17388	0.0346	0	0	0	1
6	3	4	0.06701	0.17103	0.0128	0	0	0	1
7	4	5	0.01335	0.04211	0	0	0	0	1
8	4	7	0	0.20912	0	0	0.978	0	1
9	4	9	0	0.55618	0	0	0.969	0	1
10	5	6	0	0.25202	0	0	0.932	0	1
11	6	11	0.09498	0.1989	0	0	0	0	1
12	6	12	0.12291	0.25581	0	0	0	0	1
13	6	13	0.06615	0.13027	0	0	0	0	1
14	7	8	0	0.17615	0	0	0	0	1
15	7	9	0	0.11001	0	0	0	0	1
16	9	10	0.03181	0.0845	0	0	0	0	1
17	9	14	0.12711	0.27038	0	0	0	0	1
18	10	11	0.08205	0.19207	0	0	0	0	1
19	12	13	0.22092	0.19988	0	0	0	0	1
20	13	14	0.17093	0.34802	0	0	0	0	1

For a branch $i \rightarrow j$, $z_{ij} = r_{ij} + jx_{ij}$, $y_{ij} = 1/z_{ij}$, and the charging susceptance is divided equally between the terminal shunts. Tap magnitude τ and phase shift ϕ enter the off-nominal transformer terms. These contributions assemble the complex network matrix Y .

3 Common frames, signs, and per-unit conversion

All devices use generator sign: positive complex power is injected into the network,

$$S = P + jQ = VI^*, \quad I_{bus} = \sum_{d \in bus} I_d.$$

For an IBR whose local angle is δ_c , the network-to-device transform is

$$v_d = V_x \cos \delta_c + V_y \sin \delta_c, \quad v_q = -V_x \sin \delta_c + V_y \cos \delta_c, \quad (1)$$

$$i_x = i_d \cos \delta_c - i_q \sin \delta_c, \quad i_y = i_d \sin \delta_c + i_q \cos \delta_c. \quad (2)$$

Thus $v_q = 0$ when a PLL is locked to the positive-sequence bus voltage. With this convention,

$$P_{inv} = v_d i_d + v_q i_q, \quad Q_{inv} = v_q i_d - v_d i_q.$$

If S_{base} is the system base and M_{base} is the converter base,

$$\kappa = \frac{S_{base}}{M_{base}}, \quad P_{inv} = \kappa P_{sys}, \quad Q_{inv} = \kappa Q_{sys}, \quad I_{sys} = \frac{I_{inv}}{\kappa}.$$

IBR2 uses $M_{base} = 140$ MVA, hence $\kappa = 100/140$; IBR3, IBR6, and IBR8 use 100 MVA and $\kappa = 1$.

4 Composite network DAE

Every device supplies a differential right-hand side and a positive complex current injection. The composite equations are

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{y}, \mathbf{u}; \sigma), \quad (3)$$

$$\mathbf{0} = \mathbf{g}(\mathbf{x}, \mathbf{y}, \mathbf{u}; \sigma) = \mathcal{R}(Y(\sigma)V - I_{bus}(\mathbf{x}, \mathbf{y}, \mathbf{u}; \sigma)), \quad (4)$$

where $\mathcal{R}(z) = [\Re z_1, \Im z_1, \dots]^\top$ and σ contains topology, online status, and device mode. At a declared reference bus, the appropriate KCL rows are replaced by explicit voltage constraints; the replacement is metadata-driven, not inferred inside a device.

5 Synchronous generator: EMF6 equations

5.1 State order and meaning

$$\mathbf{x}_{SG} = [\delta, \omega, E'_q, E'_d, E''_q, E''_d]^\top, \quad \mathbf{u}_{SG} = [T_m, E_{fd}]^\top.$$

Local	State	Unit	Physical meaning
1	δ	rad	electrical rotor angle in the network reference frame
2	ω	pu	rotor-speed deviation; actual speed is $\omega_b(1 + \omega)$
3	E'_q	pu	q-axis transient internal EMF
4	E'_d	pu	d-axis transient EMF; frozen at zero here because $T'_{q0} = 0$
5	E''_q	pu	q-axis subtransient internal EMF
6	E''_d	pu	d-axis subtransient internal EMF

The physical inventory is sixth order, but the normal SSSA contribution is five because the sourced Kods round-rotor data impose the singular limit $T'_{q0} = 0 \Rightarrow E'_d = 0$.

5.2 Stator algebraic equations and ODEs

The project Kundur transform gives (V_d, V_q) from (V_x, V_y, δ) . Defining $r_d = V_d - E'_d$, $r_q = V_q - E'_q$, and $\Delta = R_a^2 + X_d'' X_q''$, the stator current is

$$I_d = \frac{-R_a r_d - X_q'' r_q}{\Delta}, \quad I_q = \frac{X_d'' r_d - R_a r_q}{\Delta}, \quad (5)$$

$$T_e = V_d I_d + V_q I_q + R_a (I_d^2 + I_q^2). \quad (6)$$

On the system base, the connected-machine dynamics are

$$\dot{\delta} = \omega_b \omega, \quad (7)$$

$$2H_{sys}\dot{\omega} = T_m - T_e - D_{sys}\omega, \quad (8)$$

$$T'_{d0}\dot{E}'_q = E_{fd} + c_d E''_q - d_d E'_q, \quad (9)$$

$$T'_{q0}\dot{E}'_d = c_q E''_d - d_q E'_d, \quad (10)$$

$$T''_{d0}\dot{E}''_q = E'_q - E''_q - (X'_d - X''_d)I_d, \quad (11)$$

$$T''_{q0}\dot{E}''_d = E'_d - E''_d + (X'_q - X''_q)I_q. \quad (12)$$

When the SG breaker is open, $I_d = I_q = T_e = 0$. The angle and speed coast, and the four flux equations use the same formulas with their current terms removed. The retained T_m policy is therefore visible in the post-trip trajectory; it is not silently re-solved at every time step.

6 GFM-VSG REGFM_B1: 13-state equations

6.1 State order and device frame

$$\mathbf{x}_{GFM} = [\omega_m, \delta_{IT}, x_w, x_E, \delta_{PLL}, \xi_{PLL}, P_f, I_{d,f}, Q_f, V_f, I_{q,f}, \delta_{IT}^{max}, \delta_{IT}^{min}]^T. \quad (13)$$

Local	State	Meaning
1	ω_m	VSG speed deviation
2	δ_{IT}	PLL-relative VSG internal angle
3	x_w	damping washout state
4	x_E	voltage-controller integral state
5–6	δ_{PLL}, ξ_{PLL}	PLL angle and PI integral
7–11	$P_f, I_{d,f}, Q_f, V_f, I_{q,f}$	measured active power, d current, reactive power, voltage and q current filters
12–13	$\delta_{IT}^{max}, \delta_{IT}^{min}$	dynamic upper/lower current-angle bounds

The device voltage angle is

$$\delta_{VSM} = \delta_{PLL} + \text{clamp}(\delta_{IT}, \delta_{IT}^{min}, \delta_{IT}^{max}), \quad \text{clamp}(z, l, h) = \min(\max(z, l), h).$$

The clamp is an algebraic projection: an interior value is unchanged, a value below l is replaced by l , and a value above h is replaced by h . The derivative uses an outward-hold/inward-release rule rather than differentiating through an impossible outward motion.

6.2 PLL, VSG, voltage and filter dynamics

With $V_q = -V_x \sin \delta_{PLL} + V_y \cos \delta_{PLL}$,

$$\dot{\xi}_{PLL} = V_q, \quad \dot{\delta}_{PLL} = \omega_b(k_{p,PLL}V_q + k_{i,PLL}\xi_{PLL}), \quad (14)$$

$$2H\dot{\omega}_m = P_{ref,inv} - P_f - (1/m_p + D_1)\omega_m - D_2(\omega_m - x_w), \quad (15)$$

$$\dot{x}_w = \omega_D(\omega_m - x_w), \quad (16)$$

$$\dot{\delta}_{IT} = \text{conditional_hold}(\omega_b\omega_m - \dot{\delta}_{PLL}; \delta_{IT}^{min}, \delta_{IT}^{max}), \quad (17)$$

$$\dot{x}_E = \text{conditional_hold}_E(V_{ref} - V_f), \quad (18)$$

$$E_{VSM}^{raw} = V_{ref} - m_q Q_f + k_{pv}(V_{ref} - V_f) + k_{iv}x_E. \quad (19)$$

The voltage-integrator hold is active only when the voltage command is at a limit and the error would drive it farther outward. The measurement states are

$$T_{pf}\dot{P}_f = \kappa P - P_f, \quad T_{If}\dot{I}_{d,f} = \kappa I_d - I_{d,f}, \quad (20)$$

$$T_{Qf}\dot{Q}_f = \kappa Q - Q_f, \quad T_{Vf}\dot{V}_f = |V| - V_f, \quad (21)$$

$$T_{If}\dot{I}_{q,f} = \kappa I_q - I_{q,f}. \quad (22)$$

The moving angle bounds satisfy

$$\dot{\delta}_{IT}^{max} = \text{conditional_hold}(k_I(I_{d,max}^{SS} - I_{d,f}); 0, \delta_{max}), \quad (23)$$

$$\dot{\delta}_{IT}^{min} = \text{conditional_hold}(k_I(-K_e I_{d,max}^{SS} - I_{d,f}); -\delta_{max}, 0), \quad (24)$$

with the lower state frozen when its source flag is disabled.

The unsaturated network current is the voltage-behind-impedance relation

$$I_{unc} = \frac{E_{VSM} e^{j\delta_{VSM}} - V}{\kappa(R_e + jX_L)}.$$

Its magnitude is circularly limited at I_{maxF}/κ , after the sourced P/Q and voltage-limit logic. This current, not a separate SSSA surrogate, is injected into the common KCL equations.

7 GFL–RMS10: 10-state PLL-resolved equations

7.1 State order and meaning

$$\mathbf{x}_{GFL} = [\delta_{PLL}, \xi_{PLL}, P_f, Q_f, \xi_P, \xi_Q, \xi_{id}, \xi_{iq}, i_d, i_q]^T.$$

Local	State	Unit	Meaning
1	δ_{PLL}	rad	SRF-PLL angle used by every dq transform
2	ξ_{PLL}	pu s	PLL PI integral of v_q
3–4	P_f, Q_f	pu inv.	filtered measured injected powers
5–6	ξ_P, ξ_Q	pu s	outer active/reactive-power PI integrals
7–8	ξ_{id}, ξ_{iq}	pu s	inner d/q current-controller PI integrals
9–10	i_d, i_q	pu inv.	physical L-filter current states in the PLL frame

The last two rows are why this model is a genuine dynamic GFL rather than a PLL label attached to algebraic current commands.

7.2 PLL, power measurement, and exact P/Q feedforward

Let $D_V = v_d^2 + v_q^2$. The declared positive-sequence domain requires $D_V \geq V_{div,min}^2$. Within that domain,

$$\dot{\xi}_{PLL} = v_q, \quad \Delta\omega_{PLL} = k_{p,PLL}v_q + k_{i,PLL}\xi_{PLL}, \quad \dot{\delta}_{PLL} = \omega_b\Delta\omega_{PLL}, \quad (25)$$

$$T_P\dot{P}_f = P_{inv} - P_f, \quad T_Q\dot{Q}_f = Q_{inv} - Q_f. \quad (26)$$

The exact inverse of the dq power equations supplies the steady feedforward:

$$i_{d,ff} = \frac{v_d P_{ref} + v_q Q_{ref}}{D_V}, \quad i_{q,ff} = \frac{v_q P_{ref} - v_d Q_{ref}}{D_V}, \quad (27)$$

$$i_{d,r}^{raw} = k_{pP}(P_{ref} - P_f) + k_{iP}\xi_P + i_{d,ff}, \quad (28)$$

$$i_{q,r}^{raw} = -k_{pQ}(Q_{ref} - Q_f) - k_{iQ}\xi_Q + i_{q,ff}. \quad (29)$$

The minus sign in the q-axis controller follows the generator convention $Q = v_q i_d - v_d i_q$: at a locked PLL, positive reactive injection requires negative i_q .

7.3 Current priority, balanced LVRT, and outer anti-windup

In normal voltage, circular P-priority limits $(i_{d,r}^{raw}, i_{q,r}^{raw})$ to I_{max} : i_d is clipped first and i_q receives the remaining circle. During the declared balanced positive-sequence voltage dip,

$$i_{q,FRT} = \text{clamp}(K_{qv} \max(V_{dip} - |V| - d_b, 0), -I_{qh1}, I_{qh1}), \quad (30)$$

$$i_{d,LVPL}^{max} = \begin{cases} 0, & |V| \leq Z_{erox}, \\ L_{vpl1} \frac{|V| - Z_{erox}}{B_{rkpt} - Z_{erox}}, & Z_{erox} < |V| < B_{rkpt}, \\ L_{vpl1}, & |V| \geq B_{rkpt}, \end{cases} \quad (31)$$

$$|i_d| \leq \min(i_{d,LVPL}^{max}, \sqrt{I_{max}^2 - i_q^2}). \quad (32)$$

This is reactive-current priority during the voltage dip; it does not cover negative sequence or zero-voltage ride through.

After limiting, $P_{lim} = v_d i_{d,r} + v_q i_{q,r}$ and $Q_{lim} = v_q i_{d,r} - v_d i_{q,r}$. The outer integrators are

$$\dot{\xi}_P = AW_P(P_{ref} - P_f), \quad \dot{\xi}_Q = AW_Q(Q_{ref} - Q_f). \quad (33)$$

Each AW returns zero only when its limiter is active and its error would push the command farther outward; inward recovery remains active. The comparison tolerance is numerical metadata, not a tunable physical gain.

7.4 Inner current PI, voltage command, and L-filter ODE

With $e_d = i_{d,r} - i_d$, $e_q = i_{q,r} - i_q$, and $\omega_{PLL,pu} = 1 + \Delta\omega_{PLL}$,

$$v_{td}^{raw} = k_{pi} e_d + k_{ii} \xi_{id} + R_t i_d - \omega_{PLL,pu} L i_q + v_d, \quad (34)$$

$$v_{tq}^{raw} = k_{pi} e_q + k_{ii} \xi_{iq} + R_t i_q + \omega_{PLL,pu} L i_d + v_q. \quad (35)$$

The vector $[v_{td}^{raw}, v_{tq}^{raw}]$ is projected onto the modulation circle of radius $V_{t,max} = m_{max} V_{dc0}$. The current PI integrators use the same outward-hold/inward-release anti-windup rule:

$$\dot{\xi}_{id} = AW_{id}(e_d), \quad \dot{\xi}_{iq} = AW_{iq}(e_q).$$

Because L is stored as per-unit reactance, the per-unit inductance in the derivative denominator is L/ω_b :

$$\dot{i}_d = \frac{\omega_{PLL,pu} L i_q - R_t i_d + v_{td} - v_d}{L/\omega_b}, \quad (36)$$

$$\dot{i}_q = \frac{-\omega_{PLL,pu} L i_d - R_t i_q + v_{tq} - v_q}{L/\omega_b}. \quad (37)$$

Finally,

$$I_{net} = \frac{(i_d + j i_q) e^{j\delta_{PLL}}}{\kappa}, \quad S = V I_{net}^*.$$

8 How PF, equilibrium, SSSA, and TS use the equations

8.1 Power flow

PF does not integrate converter or SG states. For each non-reference bus it solves the specified active/reactive mismatch selected by the REF/PV/PQ semantics. In compact form, Newton iterations solve

$$J_{PF}(z^{(m)}) \Delta z^{(m)} = -F_{PF}(z^{(m)}), \quad z^{(m+1)} = z^{(m)} + \Delta z^{(m)},$$

until $\|F_{PF}\|_\infty$ is below tolerance. PF establishes network voltage and dispatch targets; it does not claim that a dynamic state is at equilibrium.

8.2 Mixed-device equilibrium

Equilibrium uses the full device equations and solves

$$f_A(x_0, y_0, u_0) = 0, \quad g(x_0, y_0, u_0) = 0,$$

for active states and algebraic variables. Frozen or inactive branch states remain in the inventory but not in the Newton unknown vector. GFL lock gives $v_q = 0$, $P_f = P_{ref}$, $Q_f = Q_{ref}$, and stationary current states. GFM equilibrium gives $\omega_m = 0$, locked PLL/filter states, and stationary voltage/current limits. SG equilibrium solves constant T_m , E_{fd} , which are then held in TS.

8.3 Full-state SSSA

Linearizing the same DAE at a fixed active regime gives

$$\begin{bmatrix} \Delta \dot{x} \\ 0 \end{bmatrix} = \begin{bmatrix} f_x & f_y \\ g_x & g_y \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix}, \quad A_{full} = f_x - f_y(g_y \backslash g_x).$$

The published matrix is $A = A_{full}(\mathcal{A}, \mathcal{A})$, where $\mathcal{A}$ is the metadata-owned active-state map. No eigenvalue is manually deleted. For $Av_i = \lambda_i v_i$ and $A^H u_i = \lambda_i^* u_i$, the vectors are paired and normalized so that $u_i^H v_i = 1$. Signed participation is $p_{ki} = u_{ki}^* v_{ki}$; the displayed nonnegative ranking is $\rho_{ki} = |p_{ki}| / \sum_j |p_{ji}|$. Mode number, raw eigen index, state index, and device index are separate quantities.

8.4 Time-domain simulation

Between event transactions, the implicit trapezoidal step solves

$$0 = x_{n+1} - x_n - \frac{h}{2} [f(x_n, y_n) + f(x_{n+1}, y_{n+1})], \quad (38)$$

$$0 = g(x_{n+1}, y_{n+1}; \sigma_{n+1}). \quad (39)$$

The public sampling interval is $h = 0.01$ s. A failed logical step may be bisected internally by the same equations, without changing the published sample grid or event landing time. Fault application/clearing and SG trip/return are separate atomic transactions with left/right KCL checks.

9 Normal PF and SSSA results

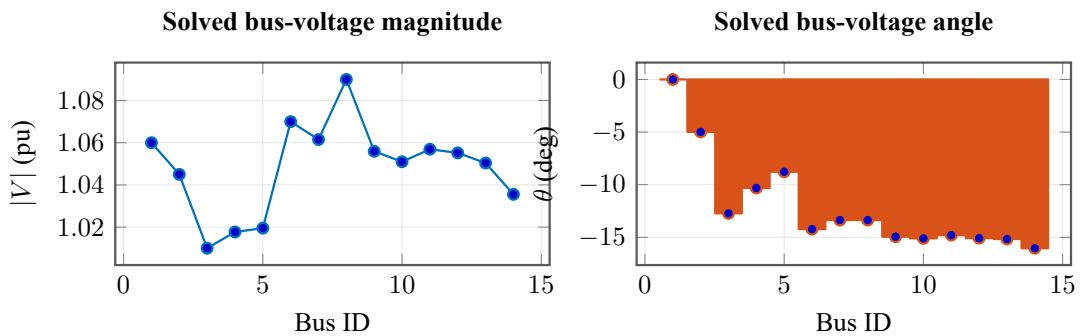


Figure 1: Normal mixed-resource power-flow voltage results.

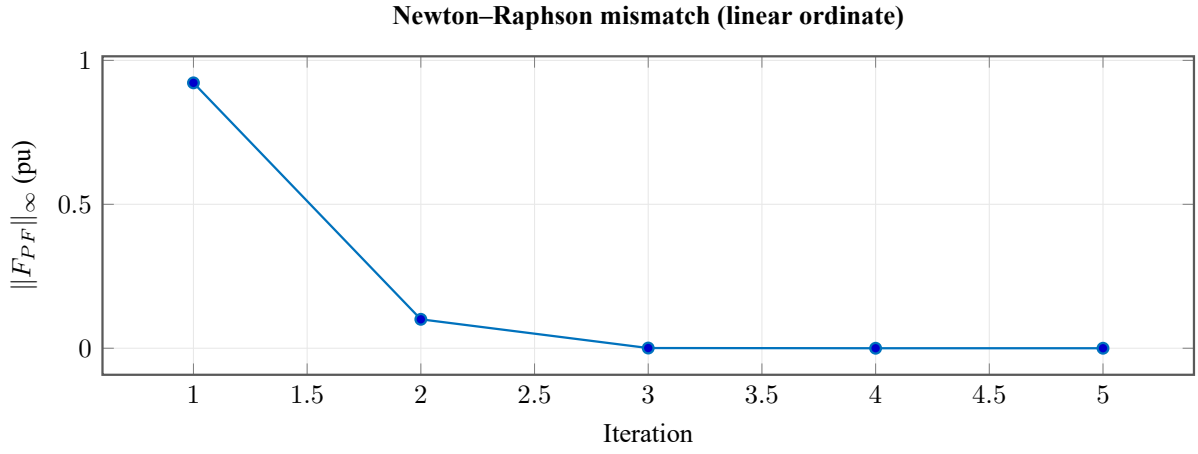


Figure 2: PF convergence on a linear, not logarithmic, vertical scale.

Table 3: Solved device injections at the normal mixed equilibrium.

Index	Device	Bus	Mode	P MW	Q MVar	$ V $ pu	$ V $ kV
1	SG1	1	sg	215.832523985106	-5.293145690365	1.06	73.14
2	IBR2	2	GFM	40	80.519244433198	1.045	72.105
3	IBR3	3	gfl	-5.23478843621713e-27	-1.2175655211094e-25	0.977765875361822	67.4658453999657
4	IBR6	6	gfl	0	0	1.02439589124011	70.6833164955677
5	IBR8	8	gfl	0	0	1.01363096475053	69.9405365677866

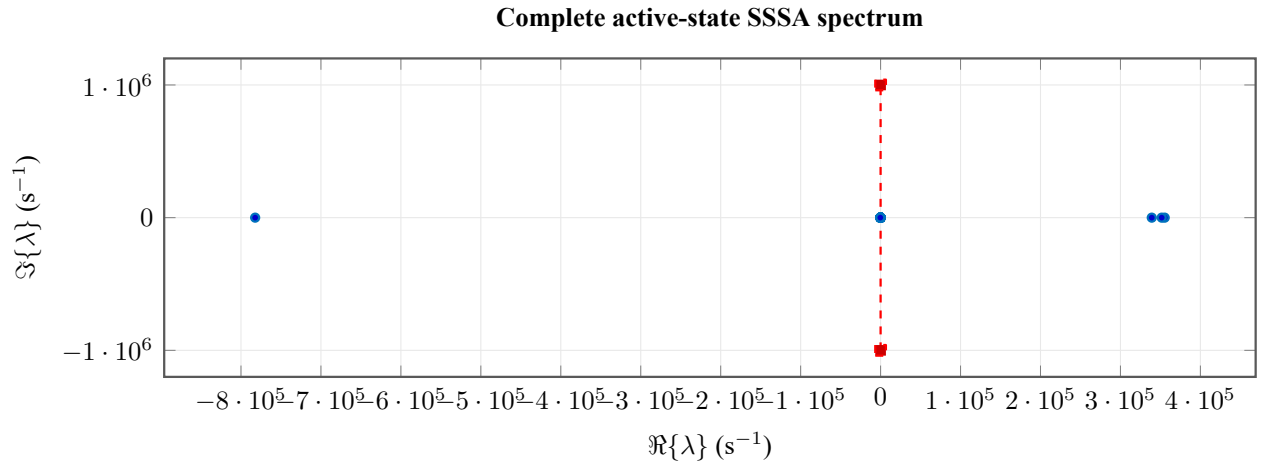


Figure 3: All 48 normal-profile eigenvalues. A positive real part is retained and classified as a physical/model result, not tuned away.

10 SG trip/return PF and SSSA comparison

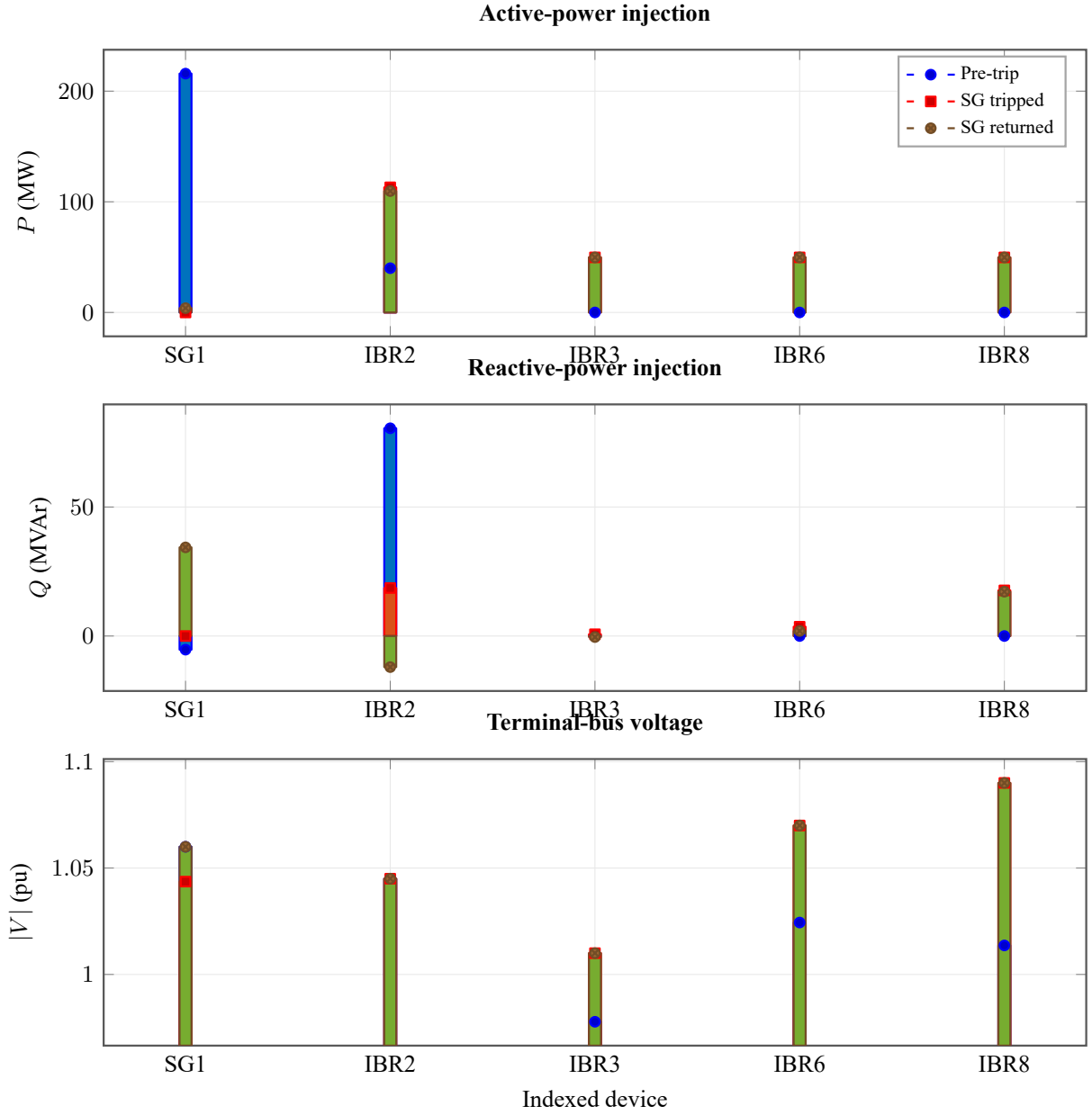


Figure 4: Symmetric native-LaTeX PF comparison. Each column is an independently solved stationary operating point.

Table 4: SSSA comparison summary. Equal row numbers do not imply matched modes.

Point	Order	$\max \Re \lambda$	Stable	Marginal	Unstable
PRE-TRIP	48	355339.094581153	43	1	4
SG-TRIPPED	52	1.29119692099525e-05	47	4	1
SG-RETURNED	57	1.27195674140501e-05	52	4	1

11 Fault-only time-domain result

The fault-only case applies a balanced positive-sequence shunt fault at bus 4 at $t = 0.10$ s and clears it at $t = 0.20$ s. No SG trip or automatic mode switch is included. The following plots use exactly the MATLAB

output series, but are typeset by $\text{\LaTeX}$.

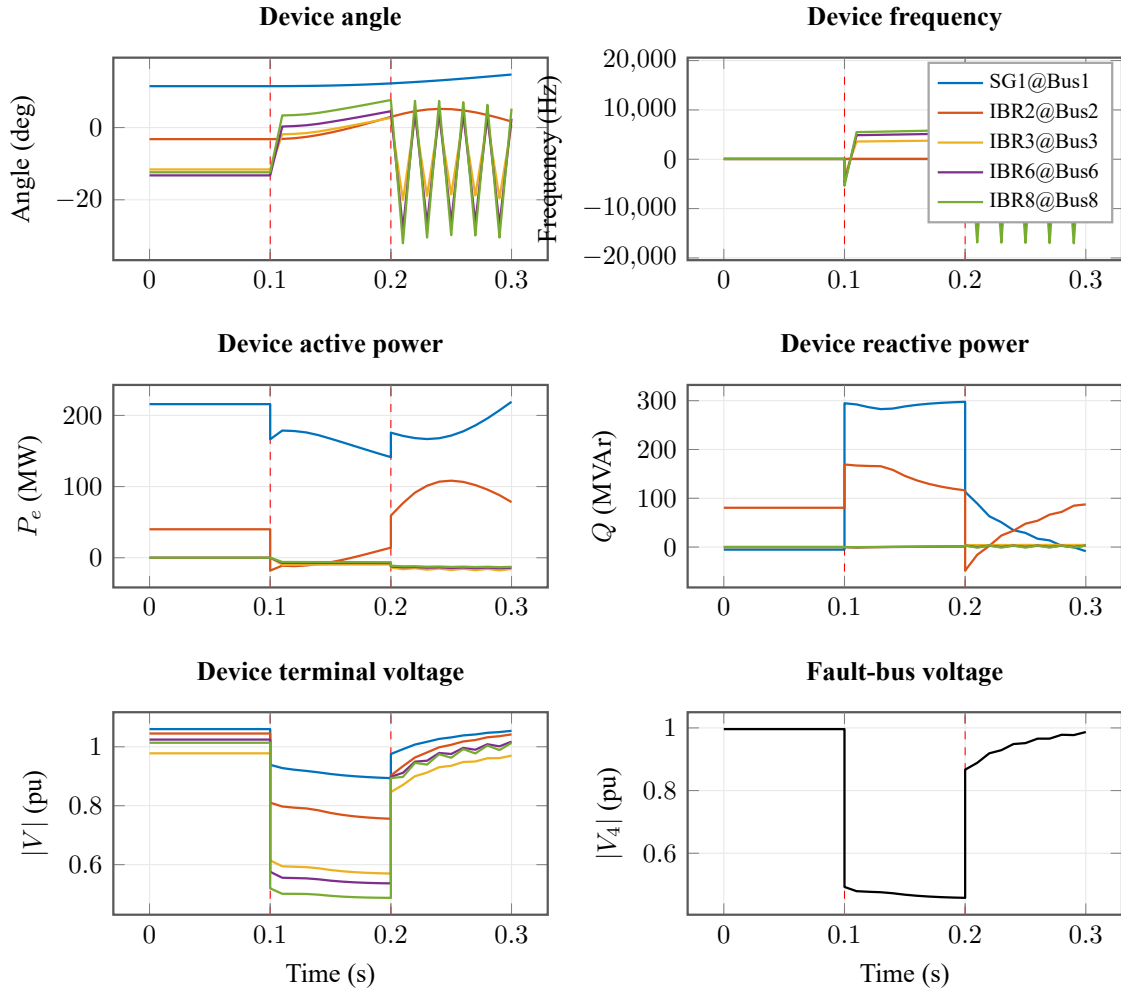


Figure 5: Fault-only TS, with identical dimensions and typography in all panels. Event times are 0.10 and 0.20 s.

12 SG-trip time-domain result

This separate run contains no fault. The SG trip request lands at $t = 0.10$ s; the automatic selector commits the post-trip GFM set and the SG injects zero current after opening. The SG-on request at $t = 0.20$ s starts the synchronism guard. If the dwell has not completed by the end of this short report window, the outcome is labelled pending rather than presented as a successful physical reclose.

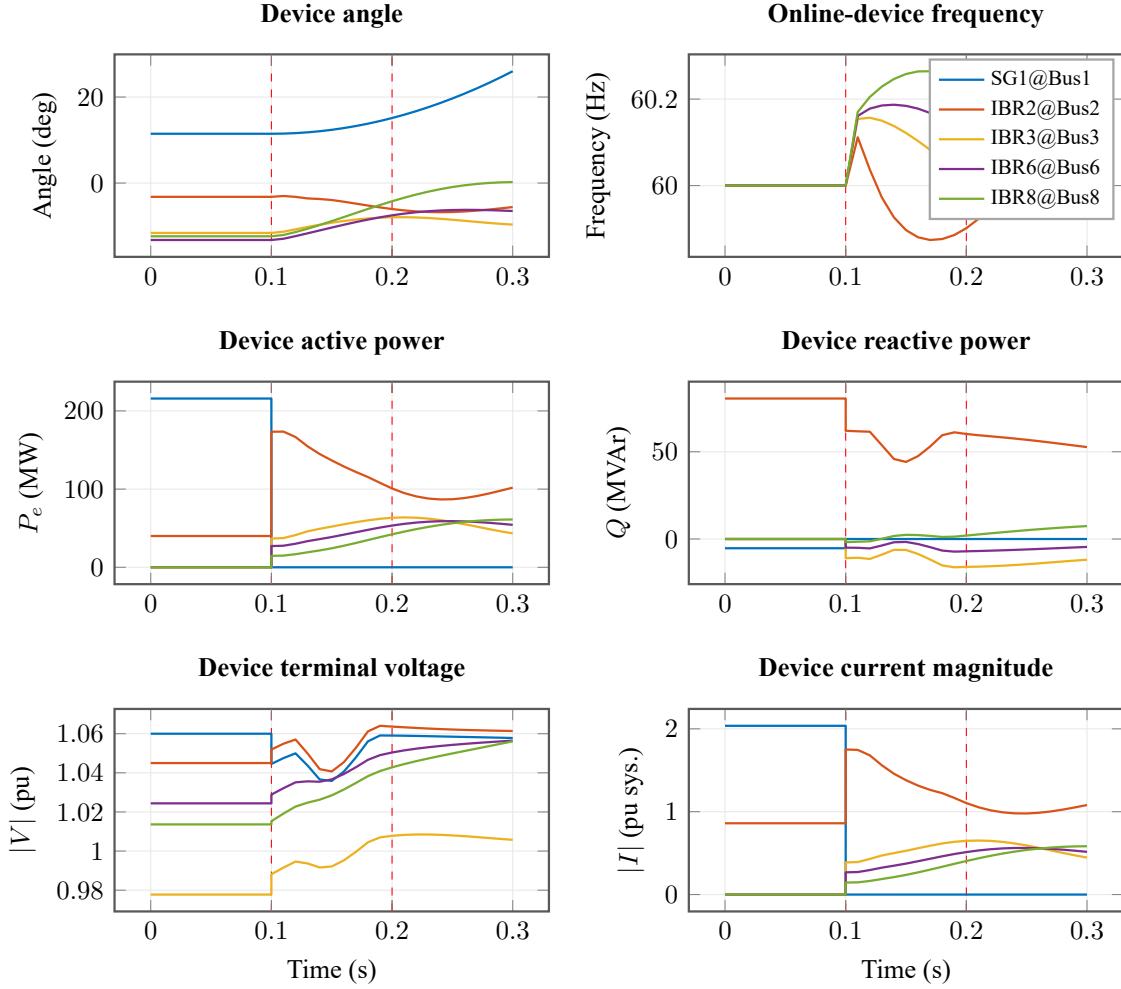


Figure 6: SG-trip/return-request TS. Missing SG frequency after the trip denotes an offline device, not zero frequency.

13 Comparison with the conventional SG analysis path

The conventional multi-SG path and the mixed IBR path share the network and analysis mathematics, but not the device states. A classical SG contributes $[\delta, \omega]$, while the present SG contributes the EMF6 inventory and the converter models contribute control and physical current states. PF is common because all resources appear as scheduled injections and voltage constraints. Equilibrium, SSSA, and TS are generic consumers of the device ABI: they do not contain a GFM-specific or GFL-specific eigenvalue matrix. Consequently, changing an IBR from GFL-RMS10 to GFM changes the active state map and the nonlinear current law, but does not change the Schur-complement or implicit-trapezoidal algorithms.

14 Limitations and interpretation

- The GFL fault contract is balanced positive-sequence RMS LVRT. It is not an EMT, negative-sequence, or zero-voltage model.
- The SG trip/return stationary comparison is not proof that a breaker reclose transaction will pass the voltage, frequency, angle and dwell guard.
- A completed numerical SSSA execution is distinct from a stable physical classification. Positive-real-part eigenvalues remain in the table and plot.
- The 13-state GFM and 10-state GFL orders need not be equal: order counts the independent memory elements required by each sourced control architecture, not the number of terminals or rated power.

15 Conclusion

The report exposes a traceable chain from case data through device equations, state order, initialization, DAE assembly, PF, SSSA, and TS. The same nonlinear SG/GFM/GFL equations initialize the equilibrium, are differentiated for SSSA, and are integrated for TS. Native $\text{\LaTeX}$ plots retain the exact MATLAB series while enforcing common typography and symmetric layout. The separated fault-only and SG-trip products prevent a combined workflow from hiding which event or model caused a response or fail-closed outcome.

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